LM-IDNet

Dirichlet–Multinomial Anomaly Detection for IoT Traffic

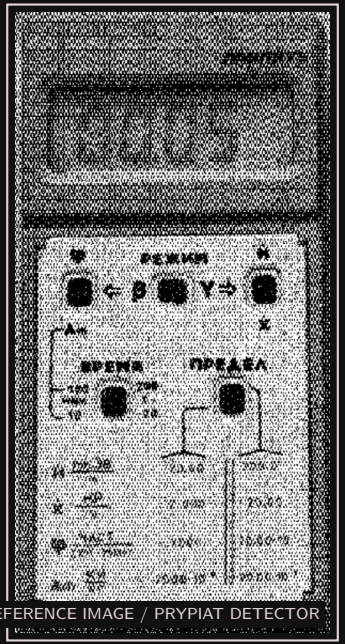
Languasco–Migliardi (LM) method

Fatemeh Mahvari

Supervisor: Professor Mauro Migliardi

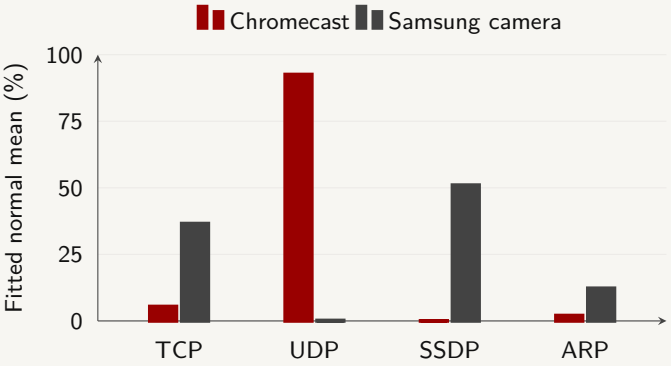
Co-supervisor: Professor Alessandro Languasco

Discussion presentation ▪ 20 October 2026



REFERENCE IMAGE / PRYPIAT DETECTOR

Device traffic profiles



Device-specific models

Normal protocol proportions differ between devices.

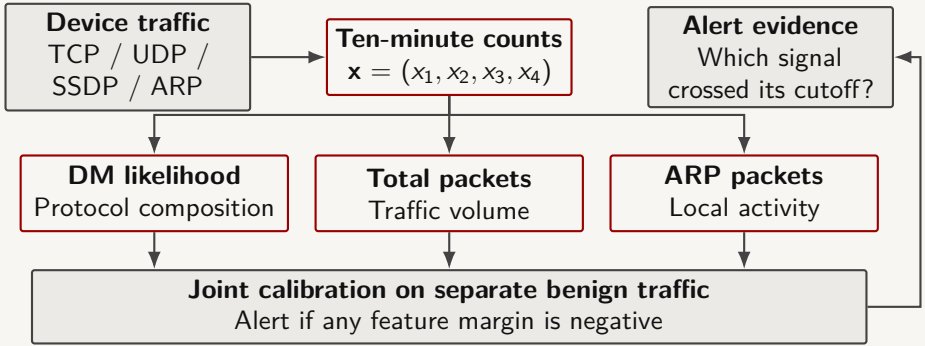
4 exclusive counts
per ten-minute window

Learn from a trusted benign period; flag unusual later behaviour.

A separate normal model captures each device's traffic pattern.

Fitted means from saved normal-only models. UDP excludes SSDP; bars are model expectations.

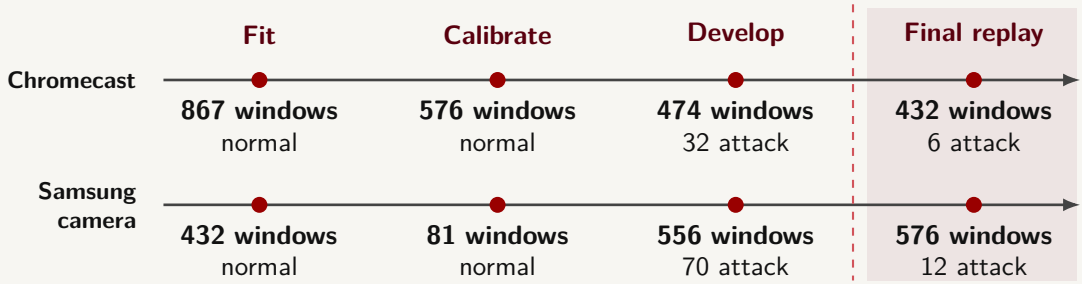
Detection model and conservative adaptation



Conservative periodic refitting: admit only sufficiently safe windows; check candidate changes before activation.

Normal-only fitting and calibration. The native LM backend evaluates the DM likelihood.

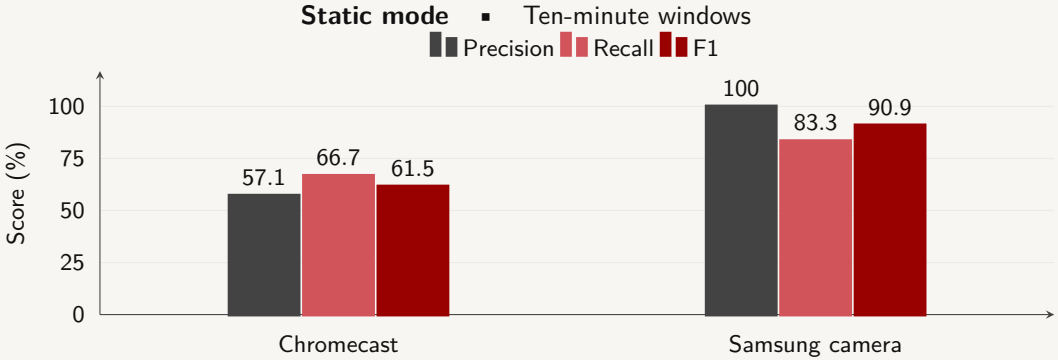
Datasets and chronological partitions



Final replay: fixed settings, saved normal-only starting pairs, labels only for metrics.

Original reference partitions. Timeline shows role order, not elapsed time. D-Link is a numerical workload without attack labels.

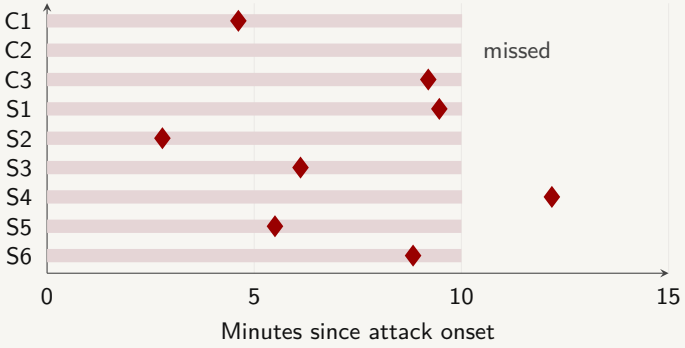
Final detection results by device



Chromecast: 4/6 attack windows flagged. Samsung camera: 10/12 flagged.

Static results. Conservative adaptive-threshold and periodic-refit modes give identical decisions; no updates were activated.

Attack coverage and detection delay



First alert
Annotated attack

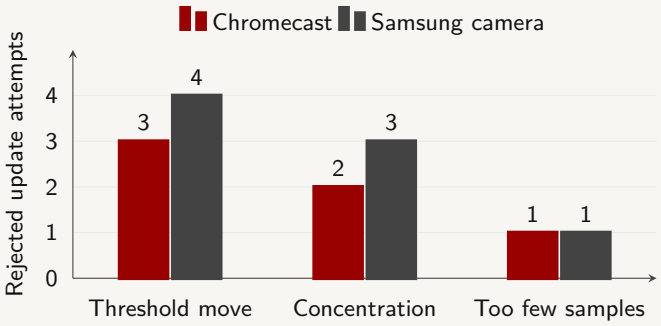
Chromecast
2/3 detected; 2/3 by end
6.9 min median

Samsung camera
6/6 detected; 5/6 by end
7.5 min median

The ten-minute collection interval is part of the detection delay.

Static mode, final official intervals; C = Chromecast, S = Samsung. Median delay is among detected attacks; alerts occur at window close.

Conservative adaptation results



Activated in each mode

0 / 3

Chromecast

0 / 4

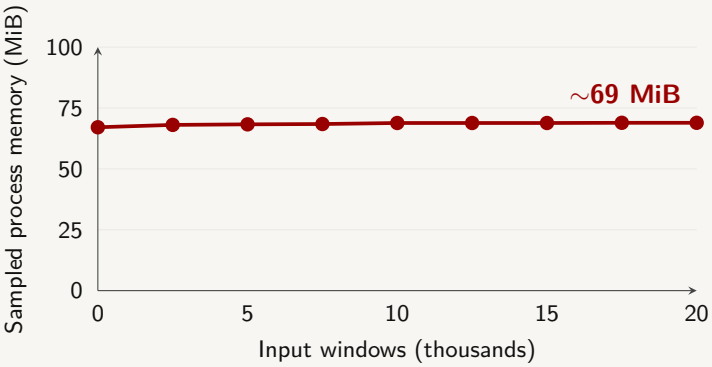
Samsung camera

Threshold-adaptive and periodic-refit modes.

No adaptive gain is established on this final replay.

Final replay. Per device: threshold attempts reject threshold movement; refits reject concentration movement or too few samples.

Edge simulation: adaptive periodic refitting



D-Link Camera 5 reference
Conservative policy
1 CPU / 512 MiB

5.079 s
worst observed processing
including a rejected refit

20,000 supplied windows
16,000 scored; 4,000 missing

Processing has observed headroom; physical gateway performance remains unmeasured.

Recorded raw-DM runtime, synthetic windows, x86_64 Docker. No update promoted; processing time excludes collection delay.

Achievements, limitations and future work

4 counts

Compact statistical core

Normal-only DM learning, native likelihood, explainable count signals.

2 devices

Final static F1

Chromecast: 61.54%
Samsung camera: 90.91%
Full decisions and update records.

$\sim$69 MiB

Operational software

Shared batch/streaming detection, guarded state, restart and rollback.

Evidence limits Only 6 final attack windows for Chromecast and 12 for Samsung; count-preserving attacks can be missed; useful conservative learning remains unproven.



Device-specific detection, explainable events, guarded adaptation, restart and rollback.

Thank you

Questions and discussion

LM-IDNet ▪ Fatemeh Mahvari

Reference slides follow: model, native method, counts, guards, window study, sources.

R1 / Model and decision rule

DM composition model $\alpha_k > 0$, $\alpha_0 = \sum_k \alpha_k$, $N = \sum_k x_k$

$$\ell(\mathbf{x}) = \log \frac{N!}{\prod_k x_k!} + \log \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + N)} + \sum_k \log \frac{\Gamma(\alpha_k + x_k)}{\Gamma(\alpha_k)}.$$

Three signals $\ell(\mathbf{x})$, $-\log(1 + N)$, $-\log(1 + x_{\mathrm{ARP}})$.

Fit-only medians and IQRs scale each signal. The minimum scaled signal is calibrated jointly at the 1% lower quantile on separate benign windows.

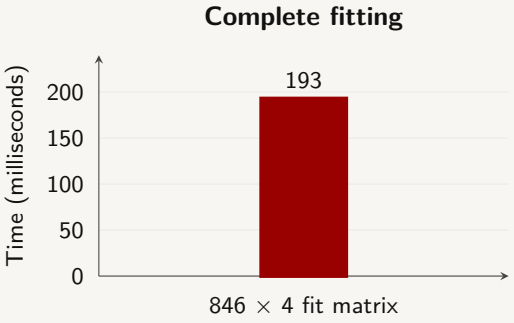
The event score is the minimum signed margin from the resulting cutoffs:

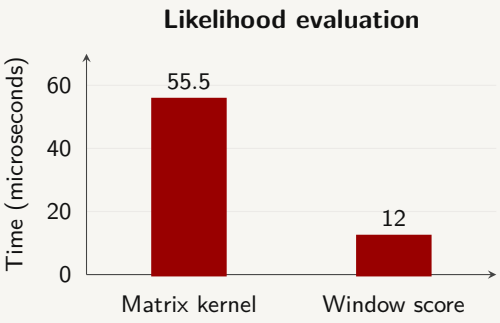
$$s = \min(m_{\mathrm{DM}}, m_N, m_{\mathrm{ARP}}), \qquad \text{alert if } s < 0, \qquad \text{severity} = \max(0, -s).$$

During updates, original DM and count references remain active detection checks.

Constant fit features use unit scale. A count-tail breach can decide early. Severity is not an attack probability.

R2 / Native Languasco–Migliardi implementation

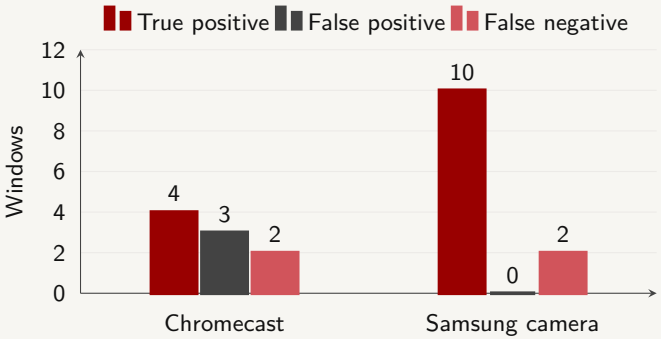




Native integration preserves the sum of independent window likelihoods.

Retained D-Link native-method timings, five batches and warm-up. Fitting excludes input/output. These are recorded measurements.

R3 / Final confusion counts by device



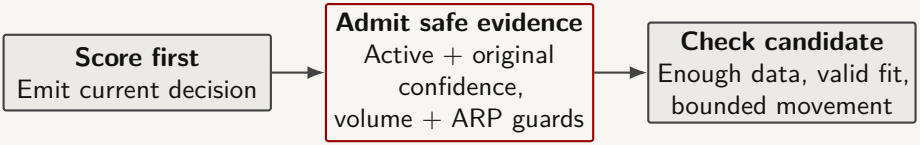
Chromecast
432 windows; 6 attack
TP = 4 FP = 3
FN = 2 TN = 423
FPR = 3/426 = 0.70%

Samsung camera
576 windows; 12 attack
TP = 10 FP = 0
FN = 2 TN = 564
FPR = 0/564 = 0%

Window recall and attack-interval coverage answer different questions.

Static mode, final partitions. A window is attack-labelled if it overlaps an official attack interval.

R4 / Conservative learning guards



$$1/1.25 \leq \alpha_0'/\alpha_{0,t} \leq 1.25 \qquad \text{concentration ratio}$$

$$\max_k |p_k' - p_{k,t}| \leq 0.05 \qquad \text{protocol proportions}$$

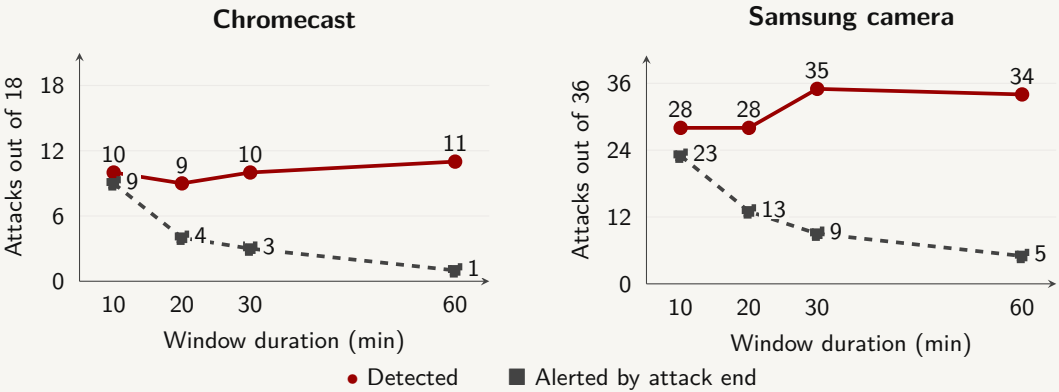
$$|\tau' - \tau_t| \leq 2\,\mathrm{IQR}_0 \qquad \text{threshold movement}$$

Defaults: 288-window buffer; attempt every 144 scored windows; 144 accepted windows required for periodic fitting.

Rejecting a candidate retains the complete active pair; accepted pairs support rollback.

Per-update guards do not establish immunity to stealthy attacks or unlimited cumulative drift. Final replay promoted no candidates.

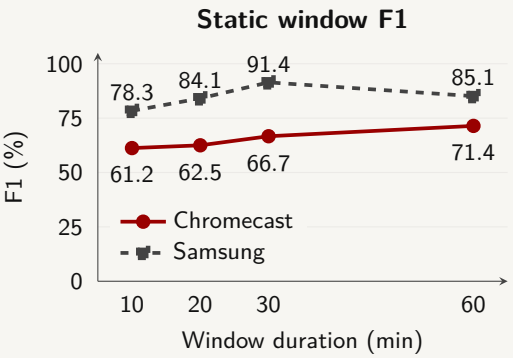
R5 / Window duration: attack coverage

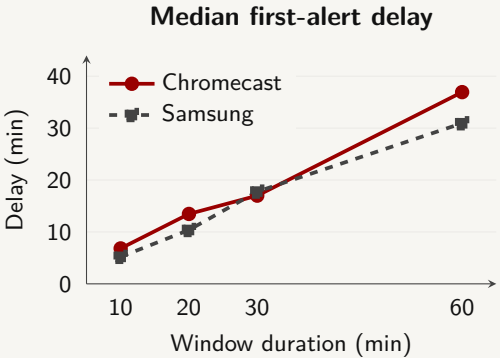


Delivered reference: ten minutes. Longer windows increase response delay.

Static development study; matched traffic within each device; fresh fitting and calibration at every duration.

R6 / Window duration: F1 and detection delay





F1 changes with label granularity; median delay is conditional on detection.

Static development study. Samsung uses May 28–29 to fit and May 30–31 to calibrate at every duration. Lines join measured settings.

R7 / References and project evidence

Statistical learning

Minka, *Estimating a Dirichlet distribution*, 2000 / revised 2012.
Fixed-point estimation of the DM parameters.

Native likelihood method

Languasco and Migliardi, *Computational Statistics* 38, 2023.
doi: 10.1007/s00180-022-01311-7
Al-Haj Baddar, Languasco and Migliardi, *IEEE TPAMI* 47(2), 2025.
doi: 10.1109/TPAMI.2024.3489645

Project evidence

Final protocol, results and events: `reports/final_evaluation/`
Duration study: `reports/window_size/`
Backend benchmark: `reports/D-LinkDayCam5/`
Supplied-window simulation: `reports/edge/`

Speaker notes map every numerical slide to its retained source files.